



Queen Mary University of London Engineering School, NPU

EXP2-4 Blamestorming

Part 2 Report

M3

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1 Results

The calculation strictly follows the procedure demonstrated in the document 'EXP2-4 Part 2 Instructions' ^[1], and the raw data comes from the Excel 'EXP2-4 Experimental Data GIC' ^[2]. The calculation results are shown in Table 1, where A, B, C, and D refer to polymer, water-solid ratio, temperature, and glass respectively, and AB refers to the interaction between A and B, so as AB, AC, AD... etc.

Table 1. The calculation results of the Effects and Random Variations in the factorial experiment.

		A	B	C	D	AB	AC	AD	BC	BD	CD
Y'	SUM+	153.42	267.44	313.7	341.76	434.36	418.26	324.36	383.88	338.14	333.32
	SUM-	513.72	399.7	353.44	325.38	232.78	248.88	342.78	283.26	329	333.82
	Diff	360.3	132.26	39.74	16.38	201.58	169.38	18.42	100.62	9.14	0.5
	Effect	45.04	16.53	4.97	2.05	25.20	21.17	2.30	12.58	1.14	0.06
R	SUM+	120.4	262.4	301.9	350.6	324.3	402	287.1	394.4	289.9	299.2
	SUM-	497.7	355.7	316.2	267.5	293.8	216.1	331	223.7	328.2	318.9
	Diff	377.3	93.3	14.3	83.1	30.5	185.9	43.9	170.7	38.3	19.7
	Effect	47.16	11.66	1.79	10.39	3.81	23.24	5.49	21.34	4.79	2.46
Random Variation: 9.66											

Effects of Y' and R respectively represent the influence on product performance and process reliability. Random variation represents the variability and reliability of outcomes in data. If the Effect value of a factor is much greater than the Random Variation value, then the factor has a real effect on the product performance or the process reliability.

2 Analysis

To identify which factors have a real effect on the product performance strength and the reliability of the process, the effects of individual factors and combined factors for Y' and R were compared with 1.5 times the Random Variation separately. The comparison results are presented in Table 1, where highlighted cells indicate real factors.

2.1 Individual factors

From Table 1, it can be concluded that A and B are individual real factors, among these A affects both product performance and process reliability, while B only affects the performance.

Regarding polymers, different types and quantities of protonated groups can influence the acidity and reactivity of the polymer. Changes in protonated groups may affect the degree of cross-linking, namely the number and density of cross-linking points ^[3]. More cross-linking points typically imply a denser polymer network, potentially enhancing the product's hardness. Additionally, different polymers have varying adaptation conditions, and conditions suitable for old polymers may not necessarily be suitable for new ones, which can alter the reliability of the process.

As for water, it plays a crucial role as a medium in this process, facilitating the progression and refinement of reactions. Therefore, changes in water content can impact the strength of the filler.

Temperature and glass, when considered individually, are not regarded as real factors.

2.2 Interactions

From Table 1, it can be inferred factors A, B, and C interact with each other to affect product performance or process reliability. Meanwhile, based on Section 2.1, an analysis was conducted to determine whether the individual factors in the interacting real factors behave the same in product performance and process reliability when combined (Table 2).

Table 2. Analysis of individual and combined factors behaviors.

Factor	Affected properties	Sameness of behaviour	
A&B	Strength	A	Same
		B	Same
A&C	Strength	A	Same
		C	Different
	Reliability	A	Same
		C	Different
B&C	Reliability	B	Different
		C	Different

A&B: Changes in the type of polymer and the content of water individually affect the strength of the filler. The interaction between A and B may result in either a synergistic effect^[4] of their individual impacts or a simple summation of the effects of A and B, but both lead to an effect on the strength of the filler.

A&C: Although temperature changes themselves do not directly affect the strength of the filler, the interaction between the polymer and temperature may alter this scenario. For example, an increase in temperature might facilitate the movement and softening of polymer chains, thereby influencing the strength and reliability of the filler. In this case, although the individual impact of temperature is not significant, its interaction with the polymer may cause a change in the strength of the filler.

B&C: While the individual effects of water and temperature do not directly alter the strength of the filler, their interaction may lead to a change in the filler's strength. An increase in water content and temperature may result in the hydrolysis of polymer chains in the filler and changes in cross-linking, thereby affecting the reliability of the process.

3 Conversation discussion

The 'blamestorming' conversation [1] will be discussed in comprehensive consideration with the analysis of the results in Section 2 to figure out the potential reasons for the process to go wrong and the correctness of the opinions.

3.1 Simon's Inference

Simon firstly says that "Wasn't that about the time we **changed the polymer**? The new polymer was introduced because it was much tougher and would allow our product to be used for restorations at the edge of the tooth.". Simon has already known that the polymer is changed for higher strength. However, he doesn't know whether the change of polymer would have any effect on the process, for instance, the new polymer may need more water to initiate the cross-linking process.

3.2 Laura's Observation

Laura says, "It doesn't mix as well – feels kind of '**sticky**'." In the mixing process, the mixture becomes "sticky" which indicates that it lacks water and is difficult for the ions to release from the glass and form ionic cross-links between the polymer chains, which in turn causes the incompleteness of the cross-linking and become the potential for the process to go wrong.

3.3 Darren's Strategy

Darren says that "Whenever I find the mix is difficult to work, I just **add a bit more water**. You can't tell the difference from our good old Fill-1-Now material." To solve the 'sticky' problem, Darren already had a solution - adding more water. However, he doesn't know if the amount of water would affect the polymer behavior on the product performance, which reveals that the company changed the formula without figuring out the insight reasons.

3.4 Anita and Simon's debate

In statement 4, Anita and Simon argue that Anita is sure that the new material's performance is **better** than the old one after substituting with the new polymer, while Simon thinks it is unreasonable because the products are tested **at room**

temperature. However, it turns out that temperature is not a single factor that affects the process. So, Simon's statement is wrong. Only when the temperature is combined with the type of polymer and the amount of water, affects the reliability of the process.

3.5 Laura's Assertion

In statement 5, Laura says "The new glass was supposed to have exactly **the same specifications** as the old one." This sentence reveals that the company directly believed the instruction from the suppliers to check the effects caused by the change of glass powder. However, the calculation results show that Glass powder is not a factor, so the statement is aligned with Laura's assertion. But this sentence implies the company's unfamiliarity with the production process.

4 Design reflection

Comparing the experimental plan proposed in Part 1 and the factorial experiment, the basic difference is that the factorial experiment is statistical method based which investigates **all the potential factors simultaneously**, while the Part 1 plan is **single variable based** instead.

As a result, the Part 1 plan does investigate each factor's impact on the product performance, however, without an explicit criterion on determining if a factor affects the outcome. But the good thing is the Part 1 plan may give some clues indicating how a factor may affect the outcome, i.e., more/less water is more conducive to the production of high-performance products, while the factorial design experiment cannot.

In addition to the problems with individual factors identification, the Part 1 plan fails to recognize the interactions between the factors. Again, without explicit statistical criteria, interactions between factors may be noticed, but further research is required to confirm which combinations affect the outcome.

Regarding the time and resources, the Part 1 plan tests two outcomes instead of one outcome in a factorial experiment, with each run repeated 3 times. Therefore, the task load of the Part 1 plan is a little bit heavier than that of the factorial experiment, while the results that can be obtained are limited.

5 Future plan

1. Customer notification and product recalling

Firstly, customers of the batch should be contacted, warned, and given apologies for potential problems with the product, and if necessary, the last batch should be recalled to avoid more serious losses, which prevents customers from losing trust in the company.

2. Further research on interactions between individual factors

The previous factorial experiment only set 2 levels for each factor and is qualitative instead of quantitative. Also, only combinations of two factors are considered, lacking in-depth analysis of the correlation between more factors. In view of the limitations of the previous factorial experiment, it is recommended that more **quantitative levels** of more individual factors identified should be set for optimal choice determination, and combinations of these factors should be taken into consideration to explore the essence of the existing problem in the process. Additionally, more outcomes, i.e., mechanical properties, are recommended to be involved, which gives more comprehensive representation of the product performance.

3. Quality control system optimization

On the one side, the staff in the company should be clearer and more familiar with the properties of the materials purchased from the suppliers. That is, they should test the impact of new samples used for production in the process, not just follow the instructions given by the suppliers. On the other hand, the R&D personnel should deeply understand the mechanism for the processing of the product instead of just making changes based on phenomena (i.e. "sticky" so add more water), and the Quality Manager should carefully save and record the important documents and avoid loss.

6 References

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